August 27, 2025

AAE 537
Hypersonic Propulsion
Fall 2025

Course Information

Objectives

The objective of this course is to advance students' knowledge of propulsion system design for missions requiring sustained flight at speeds significantly greater than the speed of sound within Earth's atmosphere. The course begins with a foundation in first principles, establishing the connection between mission requirements, trajectory design, and propulsion system performance. A range of engine cycles will be examined, with detailed evaluation of their suitability and limitations in the hypersonic regime. Technical coverage will include the major elements of high-speed propulsion systems, with emphasis on fundamental design methodologies, performance analysis, and integration considerations. Current research directions, prevailing design practices, and recognized technological limitations will also be addressed to provide context for contemporary challenges in the field. Upon completion, students will be able to perform first-order design analyses, specify propulsion system architectures appropriate for hypersonic applications, and assess their performance across a representative range of flight Mach numbers.

Prerequisites

This course requires strong undergraduate preparation in the thermal sciences; including thermodynamics, compressible aerodynamics, and heat transfer. Students in the class are also required to have prior exposure to material on engines or propulsion systems. Students should also be experienced with writing computer codes to perform engineering analysis. Remedial materials can be suggested for those who need to catch up on any of these topics.

Registration

This course has three cross-listed sections, each with an equivalent three credit hour value. The respective meeting times and/or delivery modes are as follows:

CRN 25619: In-person, on-campus
CRN 29169: Asynchronous Online Learning, West Lafayette campus
CRN 26313: Asynchronous Online Learning, Engineering Professional Education

Textbooks

There is no required text for the class but numerous papers/materials will be shared on course site. The reference books below are all available through the Purdue library (Aerospace Research Central) and serve as some general background material:

Heiser and Pratt, *Hypersonic Airbreathing Propulsion*, 1994, ISBN 978-156347-035-7
Curran and Murthy, *Scramjet Propulsion*, 2001, ISBN 978-1-60086-660-9
Murthy, *Developments in High-Speed Propulsion*, 1996, ISBN 978-1-60086-640-1
Murthy and Curran, *High-Speed Flight Propulsion Systems*, 1991, ISBN 978-1-60086-610-4
Segal, *The Scramjet Engine*, Cambridge University Press, 2009, ISBN 978-052183-815-3
Hill and Peterson, *Mechanics and Thermodynamics of Propulsion*, 1992, ISBN 0-201-14659-2

Websites

(1) For course administration, announcements, lecture notes, lecture videos, and grades: Brightspace (all sections), *Fall 2025 AAE 537 - Merge*

(2) For assignments:
Gradescope, Course ID 1086026, *AAE 537*

Course Notes

Structured notes to guide the lectures will be posted on brightspace prior to the scheduled on-campus lecture.

Grading

Final grades will be assigned based upon the homework assignments, midterm exam, and team final project with the following weights:

Homework Assignments:	20%
Midterm Exam 1:	25%
Midterm Exam 2:	25%
Final Project:	30%

In general, we will follow the 90% / 80% / 70% / 60% scale with +/- grades. Depending on class performance, the instructor reserves the right to curve the scale, if necessary, based on his discretion. The scale will never be curved downward, against the class. Those who have demonstrated an excellent understanding of airbreathing propulsion will receive an A+, A, or A- grade. Those who have demonstrated a minimum acceptable understanding will receive a C+, C, or C- grade with B grades falling in between these extremes. Those who have demonstrated an unacceptable level of understanding will receive an F grade.

Homework

Approximately five homework assignments will be given at various times during the semester. The homework problems will be posted on the course website and students are responsible for accessing them and turning in the solved problems by the noted due date. **Late homework will not be accepted, under any circumstance**. **However, your lowest homework grade will be dropped before computation of your final grade**. The class is encouraged to work together on assignments and to seek help from the TA or the professor. However, the submitted assignment must be the individual student's work. The submission must be also legible and well-organized, or points may be deducted.

Exams

There will be two midterm exams in this course. There will be no final exam for this course.
The midterm exams will be administered outside of class hours. All students enrolled in sections 25619 and 29169 must take the midterm exams in-person at the time and place noted below:

Midterm Exam 1
Midterm Exam 2

EPE students enrolled in section 26313 must coordinate their exam logistics with the EPE office and with their proctors. There are two options for exam proctoring:

(1) Use an in-person proctor through a testing center (or similar)

(2) Use an online proctoring service (e.g. Examity)

Information concerning each option can be found through Brightspace. For additional support in coordinating your exam proctoring, please contact Purdue Online Exams (purdueonlineexams@purdue.edu).

Students enrolled in sections 25619 and 29169 must take the exam on the scheduled date, at the scheduled time. Students enrolled in sections 26313 must take the exam within a 26-hour window prior-to and including the schedule time for the on-campus exam. With the exception of an emergency, failure to take the exam within the scheduled time will result in a grade of zero for that exam.
If you have a foreknown schedule conflict, then contact the professor as soon as possible. If you require accommodated testing, you are encouraged to contact the Purdue Testing Center as soon as possible to ensure your accommodation requests are fulfilled.

Project

The final project will be completed with an assigned team. Team grades will be assigned based on instructor evaluation of accuracy and completeness of material contained in a written report and a final presentation. Individual grades will be adjusted, relative to the team grade, based on feedback from peer evaluations submitted with the final report.

Other Notes and Policies

Questions

You are encouraged to attend office hours as your first source of assistance in the course. The TAs and the instructor will host several offerings throughout the week. Additional assistance will also be provided by the TAs via email. Include *AAE 537* as the start of your email subject line to ensure emails are identified and prioritized. Monitor Brightspace closely for announcements of general interest. Email with the instructor should be restricted to communication regarding individual matters course administration and similar concerns.

Changes and Emergencies

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructor via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Copyright

Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

Attendance

Recorded lecture videos will be made available to the on-campus class due to the co-listed distance section. In-person participation is still heavily encouraged. The Purdue handbook sets the following general expectation:
"The University expects that students will attend classes for which they are registered. At times, however, either anticipated or unanticipated absences can occur. The student bears the responsibility of informing the instructor in a timely fashion, when possible. The instructor bears the responsibility of trying to accommodate the student either by excusing the student or allowing the student to make up work, when possible. The University expects both students and their instructors to approach problems with class attendance in a manner that is reasonable."

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Link to Purdue's nondiscrimination policy statement.

Accessibility and Accommodations

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let us know so that we can discuss options. We also encourage you to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health

If you find yourself beginning to feel some stress, anxiety and/or feeling a bit overwhelmed, try WellTrack, https://purdue.welltrack.com/. Sign in and find information and tools at your fingertips, available to you at any time. If you need support and information about options and resources, please see the Office of the Dean of Students, http://www.purdue.edu/odos, for drop-in hours (M-F, 8 am - 5 pm). If you are struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.